NON-OBTUSE LATTICE TRIANGLES WITH NO SIDE SHORTER THAN $\sqrt{n}$ HAVE DOUBLED AREA GREATER THAN $\frac{\sqrt{3}}{2}n$: WALDMANN'S INEQUALITY FOR OEIS A344710

MATH0

Abstract. For $n \geq 1$ let $a(n)$ be twice the smallest area of a non-obtuse triangle with vertices in $\mathbb{Z}^2$ and no side shorter than $\sqrt{n}$; this is the sequence A344710 of the OEIS. We prove that $a(n) > \sqrt{3/4}\,n$ for every $n \geq 1$, the inequality recorded as conjectured in the entry by J. F. Waldmann (2021). The proof combines the identity $16\,\mathrm{area}^2 = 2XY+2YZ+2ZX-X^2-Y^2-Z^2$ for the squared side lengths X, Y, Z with the bound $(X+Y-Z)^2 \leq XY$, valid when Z is the largest of them and the triangle is non-obtuse, and it excludes equality by the irrationality of $\sqrt{3}$. The result is not new: it follows directly from the formula area $= \frac{1}{2}bc\sin\alpha$, and in the equivalent form $S(D) > \frac{\sqrt{3}}{2}D^2$ it is stated, and called obvious, in a 2019 paper of Mazel, Stuhl and Suhov on the hard-core model on $\mathbb{Z}^2$; the OEIS entry still lists it as a conjecture. Our contribution is, to our knowledge, the first formal proof: it is verified in Lean 4 with Mathlib against a statement that math0 formalized from the OEIS entry, reviewed for fidelity and frozen before any proof attempt.

1. Introduction

The OEIS entry A344710 [7], created by Jonathan F. Waldmann on May 26, 2021, has the name

> a(n)/2 is the smallest possible area of a non-obtuse triangle with coordinates in Z^2 and no side shorter than sqrt(n).

It has offset 1 and lists the 49 terms

$$1, 2, 4, 4, 5, 8, 8, 8, 9, 10, 12, 12, 12, 15, 15, 15, 15, 18, 20, 20, 23, 23, 23, 23, 23, 24, 28, \ldots, 45$$

for $n = 1, \ldots, 49$. For example, $a(4) = 4$, attained by the right isosceles triangle with vertices $(0,0), (2,0), (0,2)$. Its FORMULA section contains the line "`a(n) > sqrt(3/4)*n (conjectured).`", and the companion entry A344845 [8] (same author, May 29, 2021), in which the shortest side has length exactly $\sqrt{m}$ for m a sum of two squares [6], records the same inequality as conjectured. An equilateral triangle of side $\sqrt{n}$ has doubled area $\frac{\sqrt{3}}{2}n = \sqrt{3/4}\,n$, so the conjecture says that lattice triangles cannot do as well as the equilateral one. We prove it (Theorem 2).

Prior work and status. We do not claim the result as new. For a non-obtuse triangle with largest angle γ and adjacent sides $b, c \geq \sqrt{n}$ one has $\pi/3 \leq \gamma \leq \pi/2$, hence $2\,\mathrm{area} = bc\sin\gamma \geq \frac{\sqrt{3}}{2}n$, and equality would require an equilateral triangle, which cannot have its vertices in $\mathbb{Z}^2$. This is the classical argument; our proof (Section 3) is an algebraic form of it, see Remark 8.

The quantity $a(n)$ itself was studied before the OEIS entry was created. Mazel, Stuhl and Suhov [4], in their work on the high-density hard-core model on $\mathbb{Z}^2$ with Euclidean exclusion distance D (arXiv, first version September 2019), consider the problem of minimizing the area of a $\mathbb{Z}^2$-triangle with side lengths at least D and angles at most $\pi/2$ [4, (2.1)], and write $S(D)$

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for twice the minimal area; they assume throughout that D^2 is a sum of two squares. For such $n = D^2$ one therefore has $a(n) = S(\sqrt{n})$. This loses nothing: squared distances in $\mathbb{Z}^2$ are sums of two squares, so $a(n) = a(n')$ for the least sum of two squares $n' \geq n$ (cf. the second comment of A344710), and a lower bound $\frac{\sqrt{3}}{2}n'$ for $a(n')$ gives the bound $\frac{\sqrt{3}}{2}n$ for $a(n)$. Their Lemma 3.4.4 states

$$\tfrac{\sqrt{3}}{2}D^2 < S(D) < \tfrac{\sqrt{3}}{2}D^2 + \sqrt{2}\,D,$$

and its proof says "The lower bound for $S(D)$ is obvious." Krachun [2] defines the same $S(D)$ for every $D > 0$ [2, Definition 2.1], so that $a(n) = S(\sqrt{n})$ for all $n \geq 1$, and proves $S(D) = \frac{\sqrt{3}}{2}D^2 + O(D^{4/5})$ [2, Theorem 2]. Thus $a(n) = \frac{\sqrt{3}}{2}n + O(n^{2/5})$, and the constant $\frac{\sqrt{3}}{2}$ in Theorem 2 cannot be improved. Mazel, Stuhl and Suhov also solve the disk-packing problem on $\mathbb{Z}^2$ for disks of diameter D (their Lemma 3.5 gives the maximal density $1/S(D)$); this is the packing problem behind the sequence $b(n)$ discussed in the comments of A344710, which we do not treat here. Neither paper is cited in A344710 or A344845, and both entries (revisions #84 and #52, February 16, 2022) still mark the inequality as conjectured. It is not among the statements of the formal-conjectures repository [1] at the commit pinned by math0 (`c77fdd4`). We therefore classify the result as the first formal proof, to our knowledge, of a known result.

Method. Write X, Y, Z for the squared side lengths and D for twice the signed area. The Lagrange identity gives $4D^2 = 4XY - (X + Y - Z)^2$; if Z is the largest of the three and the triangle is non-obtuse then $0 \leq X + Y - Z \leq \min(X, Y)$, so $4D^2 \geq 3XY \geq 3n^2$ (Lemmas 3–5). Equality $4D^2 = 3n^2$ with $n \geq 1$ would make $\sqrt{3}$ rational (Lemma 7).

2. Statement

For $A, B, C \in \mathbb{Z}^2$ and $n \in \mathbb{N}$ put

$$\det_2(A, B, C) = \det(B - A,\, C - A), \qquad |PQ|^2 = (P_1 - Q_1)^2 + (P_2 - Q_2)^2,$$

so that $|\det_2(A, B, C)|$ is twice the area of the triangle ABC. The angle of ABC at A is at most $\pi/2$ if and only if $(B - A) \cdot (C - A) \geq 0$, and similarly at B and C.

Definition 1. A triple (A, B, C) of points of $\mathbb{Z}^2$ is *n-admissible* if $\det_2(A, B, C) \neq 0$, the three squared side lengths $|AB|^2, |BC|^2, |CA|^2$ are at least n, and

$$(B - A) \cdot (C - A) \geq 0, \qquad (C - B) \cdot (A - B) \geq 0, \qquad (A - C) \cdot (B - C) \geq 0.$$

Let $a(n)$ be the least value of $|\det_2(A, B, C)|$ over all n-admissible triples.

The right isosceles triangle $(0, 0), (n + 1, 0), (0, n + 1)$ is n-admissible, so the minimum exists. Thus $a(n)/2$ is the smallest area of a non-obtuse lattice triangle with no side shorter than $\sqrt{n}$, which is the definition of A344710.

Theorem 2. *For every integer $n \geq 1$,*

$$a(n) > \sqrt{3/4}\, n = \tfrac{\sqrt{3}}{2}\, n, \qquad \textit{equivalently} \qquad 4a(n)^2 > 3n^2.$$

The conjecture appears in the FORMULA section of A344710 [7] as the line

```
a(n) > sqrt(3/4)*n (conjectured).
```

This line is unsigned; the entry's author is Jonathan F. Waldmann (May 26, 2021), and its last revision (#84) is dated February 16, 2022.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A344710.lean`, reviewed for fidelity and frozen (SHA-256 `4f0e6940...6fdf74`, recorded in `lean/Math0/Statements/FROZEN.sha256`) before any proof attempt. Its definitions read as follows (verbatim; the copyright header, the import, docstrings, blank lines, the decidability instance, the bounded search used for the data check and all theorems are omitted):

```
namespace Math0.Statements.A344710
def det2 (A B C : ℤ × ℤ) : ℤ := (B.1 - A.1) * (C.2 - A.2) - (C.1 - A.1) * (B.2 - A.2)
def dist2 (P Q : ℤ × ℤ) : ℤ := (P.1 - Q.1) * (P.1 - Q.1) + (P.2 - Q.2) * (P.2 - Q.2)
def dotAt (V P Q : ℤ × ℤ) : ℤ := (P.1 - V.1) * (Q.1 - V.1) + (P.2 - V.2) * (Q.2 - V.2)
def IsValid (n : ℕ) (A B C : ℤ × ℤ) : Prop :=
  det2 A B C ≠ 0 ∧
  (n : ℤ) ≤ dist2 A B ∧ (n : ℤ) ≤ dist2 B C ∧ (n : ℤ) ≤ dist2 C A ∧
  0 ≤ dotAt A B C ∧ 0 ≤ dotAt B C A ∧ 0 ≤ dotAt C A B
def areas (n : ℕ) : Set ℕ :=
  {k | ∃ A B C : ℤ × ℤ, IsValid n A B C ∧ (det2 A B C).natAbs = k}
noncomputable def a (n : ℕ) : ℕ := sInf (areas n)
def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n → Real.sqrt (3 / 4) * (n : ℝ) < (a n : ℝ)
end Math0.Statements.A344710
```

Here `det2`, `dist2` and `dotAt V P Q` $= (P-V)\cdot(Q-V)$ are $\det_2$, the squared distance and the dot product at a vertex, and `IsValid` is Definition 1. `sInf` on $\mathbb{N}$ is the least element of a nonempty set (and 0 for the empty set), so the junk value cannot occur once `areas` n is shown to be nonempty. The file proves, among others (statements shown, proofs omitted):

```
theorem nonObtuse_iff (A B C : ℤ × ℤ) :
    (0 ≤ dotAt A B C ∧ 0 ≤ dotAt B C A ∧ 0 ≤ dotAt C A B) ↔
      2 * max (dist2 A B) (max (dist2 B C) (dist2 C A)) ≤
        dist2 A B + dist2 B C + dist2 C A
theorem areas_nonempty (n : ℕ) : (areas n).Nonempty
theorem isLeast_a (n : ℕ) : IsLeast (areas n) (a n)
theorem aRegion_data : (List.range 49).map (fun i => aRegion (i + 1)) = data
theorem a_data_le : ∀ i < 49, a (i + 1) ≤ data.getD i 0
```

The lemma `nonObtuse_iff` shows that the angle condition is the criterion

$$2\max\{|AB|^2,|BC|^2,|CA|^2\} \le |AB|^2+|BC|^2+|CA|^2$$

given in the first comment of the entry. The lemma `areas_nonempty` uses the right isosceles triangle with legs $n+1$, and `isLeast_a` states that `a` n is the least element of `areas` n. The non-collinearity condition $\det_2 \neq 0$ is not an extra restriction for $n \ge 1$: three collinear points that are pairwise distinct have an angle π, and two coincident points give a side of length 0. The statement is the FORMULA line with n ranging over the offset range $n \ge 1$; $3/4$ is real division, and no hypotheses are added.

Data check. Since the minimum defining `a` ranges over all triples of points of $\mathbb{Z}^2$, `a` is not computable as written. The statement file therefore defines a helper `aRegion n`, the least $|\det_2(A,B,C)|$ over n-admissible triples with $A=(0,0)$, C in the annulus $n \le |P|^2 \le \lfloor 5n/4 \rfloor + 2$ and B in the part $0 \le y \le x$ of that annulus (the search described in the comments of the entry, slightly widened). The lemma `aRegion_data`, proved by `decide +kernel` (evaluation in the kernel, which adds no axioms), states that `aRegion` reproduces all 49 listed terms, and `a_data_le` transfers this to $a(n) \le$ listed term for $n = 1,\dots,49$. The matching lower bound $a(n) \ge$ listed term would require that the search region always contains an optimal triangle; this is not proved in Lean. An independent check is given in Remark 10.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A344710.lean`; the name of the Lean declaration is given with each step. Throughout,

$$Q(X,Y,Z) = 2XY + 2YZ + 2ZX - X^2 - Y^2 - Z^2,$$

a symmetric polynomial. For a triple (A,B,C) we write

$$X = |AB|^2, \qquad Y = |CA|^2, \qquad Z = |BC|^2, \qquad D = \det_2(A,B,C).$$

Lemma 3. *For all $A,B,C \in \mathbb{Z}^2$,*

$$2(B-A)\cdot(C-A) = X + Y - Z, \qquad 2(C-B)\cdot(A-B) = X + Z - Y,$$
$$2(A-C)\cdot(B-C) = Y + Z - X, \qquad 4D^2 = Q(X,Y,Z).$$

Proof. Put $u = B-A$ and $v = C-A$. Then $X = |u|^2$, $Y = |v|^2$ and $Z = |u-v|^2 = X+Y-2\,u\cdot v$, which is the first identity; the other two follow in the same way at the vertices B and C. With $D = \det(u,v)$, the Lagrange identity $|u|^2|v|^2 = (u\cdot v)^2 + \det(u,v)^2$ gives

$$4D^2 = 4XY - (2\,u\cdot v)^2 = 4XY - (X+Y-Z)^2 = Q(X,Y,Z).$$

In Lean all four identities are polynomial identities in the six coordinates and are proved by `ring` (inside `det_sq_ge`). □

Lemma 4 (`heron_bound_of_max`)**.** *Let $X,Y,Z,n \in \mathbb{Z}$ with $0 \le n \le X$, $n \le Y$, $0 \le X+Y-Z$, $X \le Z$ and $Y \le Z$. Then $Q(X,Y,Z) \ge 3n^2$.*

Proof. Put $s = X+Y-Z$. Then $0 \le s$, $s \le X$ (because $Y \le Z$) and $s \le Y$ (because $X \le Z$). Since $0 \le s \le X$ and $0 \le s \le Y$, we get $s^2 \le XY$. Also $0 \le n \le X$ and $n \le Y$ give $n^2 \le XY$. Expanding,

$$Q(X,Y,Z) = 4XY - s^2 \ge 3XY \ge 3n^2.$$

In Lean the two products are bounded by `mul_le_mul`, and `nlinarith` combines them: $Q - 3n^2 = (XY - s^2) + 3(XY - n^2)$. (The intermediate bound $Q \ge 3XY$ is not stated separately in Lean.) □

Lemma 5 (`heron_bound`)**.** *Let $X,Y,Z,n \in \mathbb{Z}$ with $0 \le n$, $n \le X$, $n \le Y$, $n \le Z$ and*

$$0 \le X+Y-Z, \qquad 0 \le X+Z-Y, \qquad 0 \le Y+Z-X.$$

Then $Q(X,Y,Z) \ge 3n^2$.

Proof. If Z is the largest of X,Y,Z, apply Lemma 4 to (X,Y,Z). If Y is the largest, apply it to (X,Z,Y), using $0 \le X+Z-Y$; if X is the largest, apply it to (Y,Z,X), using $0 \le Y+Z-X$. Since Q is symmetric, each case gives $Q(X,Y,Z) \ge 3n^2$. In Lean the case distinction is made by nested case splits on `le_total`. □

Lemma 6 (`det_sq_ge`)**.** *If (A,B,C) is n-admissible, then $3n^2 \le 4\,\det_2(A,B,C)^2$.*

Proof. By admissibility $X,Y,Z \ge n \ge 0$, and by Lemma 3 the three dot product conditions say $X+Y-Z \ge 0$, $X+Z-Y \ge 0$ and $Y+Z-X \ge 0$. Lemma 5 gives $Q(X,Y,Z) \ge 3n^2$, and $Q(X,Y,Z) = 4D^2$ by Lemma 3. □

Lemma 7 (`four_sq_ne_three_sq`)**.** *There are no $m,n \in \mathbb{N}$ with $n \ge 1$ and $4m^2 = 3n^2$.*

Proof. Suppose $4m^2 = 3n^2$ with $n \ge 1$. Then $(2m)^2 = 3n^2$, and taking square roots in $\mathbb{R}$ gives $\sqrt{3}\,n = 2m$, so $\sqrt{3} = 2m/n \in \mathbb{Q}$. This contradicts the irrationality of $\sqrt{3}$, which Mathlib provides as `Nat.Prime.irrational_sqrt` for the prime 3. (Alternatively: $m \ne 0$ since $3n^2 > 0$, and then the 3-adic valuation of $4m^2$ is even while that of $3n^2$ is odd.) □

Proof of Theorem 2 (`result`). Let $n \geq 1$. By `isLeast_a`, $a(n)$ belongs to the set of values $|\det_2(A,B,C)|$, so there is an n-admissible triple (A,B,C) with $|\det_2(A,B,C)| = a(n)$. By Lemma 6, $3n^2 \leq 4a(n)^2$, and by Lemma 7 the two sides differ. Hence

$$3n^2 < 4a(n)^2.$$

In particular $a(n) > 0$, since otherwise $3n^2 < 0$. As $n \geq 0$ we have $\sqrt{3/4}\,n = \sqrt{3/4}\,\sqrt{n^2} = \sqrt{\frac{3}{4}n^2}$, and for $a(n) > 0$ the inequality $\sqrt{\frac{3}{4}n^2} < a(n)$ is equivalent to $\frac{3}{4}n^2 < a(n)^2$ (Mathlib's `Real.sqrt_lt'`), which we have shown. □

Remark 8 (geometric form)**.** If Z is the largest squared side, the angle γ at A, between the sides of squared lengths X and Y, is the largest angle, and $X + Y - Z = 2\sqrt{XY}\cos\gamma$. The inequality $(X + Y - Z)^2 \leq XY$ in the proof of Lemma 4 is $\cos^2\gamma \leq \frac{1}{4}$, that is, $\gamma \geq \pi/3$, which holds because γ is the largest angle; the condition $X + Y - Z \geq 0$ is $\gamma \leq \pi/2$. Then $4D^2 = 4XY\sin^2\gamma \geq 3XY$ is the classical bound $2\,\mathrm{area} = \sqrt{XY}\sin\gamma \geq \frac{\sqrt{3}}{2}\sqrt{XY}$. Equality in Lemma 6 would need $X = Y = Z = n$ and $\gamma = \pi/3$, an equilateral lattice triangle.

Remark 9 (the angle condition is needed)**.** Without the non-obtuseness condition the minimum is 1 for every n: for $k \geq 1$ the triangle $(0,0),(k,1),(2k+1,2)$ has squared sides k^2+1, $(k+1)^2+1$ and $(2k+1)^2+4$, all at least n once $k^2+1 \geq n$, and $|\det_2| = |2k-(2k+1)| = 1$.

Remark 10 (a complete finite search; not formalized)**.** The proof of Lemma 4 gives more than Theorem 2: in every n-admissible triangle the squared lengths X, Y of the two sides at the largest angle satisfy $3XY \leq 4D^2$. Let $k = \lceil\sqrt{n}\,\rceil$; the triangle $(0,0),(k,0),(0,k)$ is n-admissible, so $a(n) \leq k^2$. A triangle with $|D| \leq k^2$ therefore has $X, Y \leq 4k^4/(3n)$, and after translating its largest-angle vertex to the origin, $a(n)$ is a minimum over finitely many pairs of lattice points. Running this search (as the math0 referees did independently) reproduces the 49 listed terms and gives, for $n = 50,\ldots,64$,

$$45,\ 48,\ 48,\ 52,\ 53,\ 53,\ 53,\ 53,\ 53,\ 56,\ 56,\ 56,\ 56,\ 56,\ 56.$$

A comment in A344710 lists "conjectured next terms (from the conjectured inequality)" $a(50),\ldots,a(64) = 45,48,48,52,55,55,55,55,55,56,\ldots,56$; the guesses $a(54) = \cdots = a(58) = 55$ are too large. This can be checked by hand: the triangle $(0,0),(-8,-1),(-5,6)$ has squared sides 65, 61 and 58, satisfies $2\cdot 65 \leq 65+61+58$, and has $|\det_2| = |-48-5| = 53$; since a is non-decreasing, $a(n) \leq 53$ for all $n \leq 58$. Neither the search nor these values are part of the Lean development, and the frozen statement does not formalize the guessed terms.

4. Machine verification

Files and declarations. The proof is the file `lean/Math0/Frontier/OEIS/A344710.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A344710.lean` (Section 2) and the audit commands of `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A344710.result`, of type `Math0.Statements.A344710.statement`; the auxiliary lemmas of Section 3 are private. Their statements read (verbatim; proofs and docstrings omitted):

```
private lemma heron_bound_of_max (X Y Z n : ℤ) (hn : 0 ≤ n) (hX : n ≤ X) (hY : n ≤ Y)
    (h0 : 0 ≤ X + Y - Z) (hZX : X ≤ Z) (hZY : Y ≤ Z) :
    3 * n ^ 2 ≤ 2 * X * Y + 2 * Y * Z + 2 * Z * X - X ^ 2 - Y ^ 2 - Z ^ 2
private lemma heron_bound (X Y Z n : ℤ) (hn : 0 ≤ n) (hX : n ≤ X) (hY : n ≤ Y) (hZ : n ≤ Z)
    (h1 : 0 ≤ X + Y - Z) (h2 : 0 ≤ X + Z - Y) (h3 : 0 ≤ Y + Z - X) :
    3 * n ^ 2 ≤ 2 * X * Y + 2 * Y * Z + 2 * Z * X - X ^ 2 - Y ^ 2 - Z ^ 2
private lemma det_sq_ge {n : ℕ} {A B C : ℤ × ℤ} (h : IsValid n A B C) :
    3 * (n : ℤ) ^ 2 ≤ 4 * (det2 A B C) ^ 2
```

```
private lemma four_sq_ne_three_sq (m n : ℕ) (hn : 1 ≤ n) : 4 * m ^ 2 ≠ 3 * n ^ 2
theorem result : statement
#math0_settles Math0.Frontier.A344710.result Math0.Statements.A344710.statement
```

In `det_sq_ge` the variables X, Y, Z of `heron_bound` are instantiated as `dist2 A B`, `dist2 C A` and `dist2 B C`, as in Section 3. From the statement file the proof uses only the definitions and `isLeast_a` (which rests on `areas_nonempty`); it does not use the data lemmas. Neither file contains `sorry`, `native_decide`, `axiom` or `set_option`; the data lemma `aRegion_data` is proved by `decide +kernel`.

Audit. The command `#math0_settles` (`lean/Math0/Audit.lean`), used for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (rejecting `sorryAx`, the axiom behind `native_decide`, and custom axioms); and (3) the type of `result` is syntactically `statement` or its negation, up to metadata, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A344710.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A344710.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A344710.result' proves 'Math0.Statements.A344710.statement'
```

The data lemma `aRegion_data` depends on no axioms.

Toolchain and reproduction. We use Lean 4 [5] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [1] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A344710.lean` in the directory `lean`, once the statement module has been built; this takes well under a minute on our machine, most of it spent loading Mathlib. Building the statement module itself includes the kernel evaluation of `aRegion_data`, about 23 seconds of kernel time.

References

[1] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[2] D. Krachun, *Triangles on $\mathbb{Z}^2$ and sliding phenomenon*, arXiv:1912.07566, 2019; theorem numbers as in v2 (June 2020, PDF title *Extreme Gibbs measures for high-density hard-core model on $\mathbb{Z}^2$*).

[3] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[4] A. Mazel, I. Stuhl, and Y. Suhov, *High-density hard-core model on $\mathbb{Z}^2$ and norm equations in ring $\mathbb{Z}[\sqrt[6]{-1}]$*, arXiv:1909.11648, 2019 (v3, June 2023).

[5] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[6] The OEIS Foundation Inc., `https://oeis.org/A001481`, Numbers that are the sum of 2 squares; entry by N. J. A. Sloane.

[7] The OEIS Foundation Inc., `https://oeis.org/A344710`, entry by J. F. Waldmann (May 26, 2021); revision #84, February 16, 2022.

[8] The OEIS Foundation Inc., `https://oeis.org/A344845`, entry by J. F. Waldmann (May 29, 2021); revision #52, February 16, 2022.